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Gaze smoothing filter

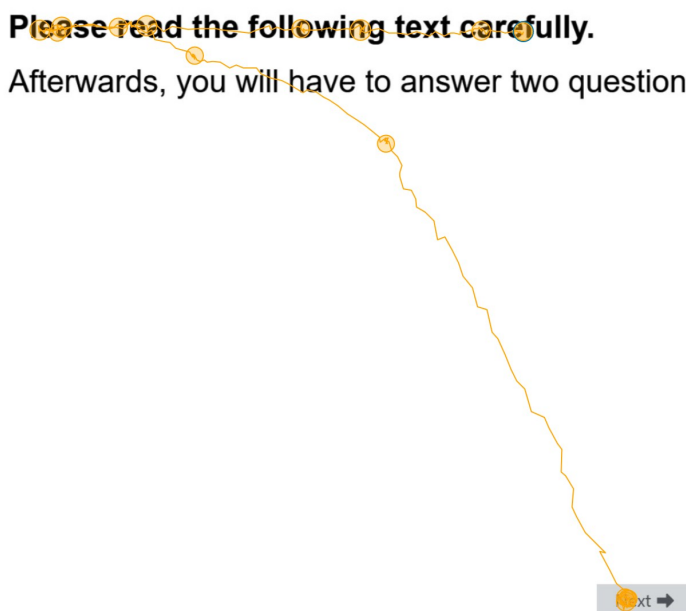
This article covers information related to the gaze smoothing filter in iMotions version 8.1 and later.

Background

As with any biosignal, the eye-tracking signal is noisy, and therefore the coordinates of the collected points are subject to a certain margin of uncertainty, depending on the accuracy and precision of the tracker used, as well as the experimental conditions. As a result, the collected eye gaze path can present jagged lines. This can be mitigated with a gaze smoothing filter, which makes the gaze path less noisy.

Please read the following text carefully.

Afterwards, you will have to answer two questions about architecture in Copenhagen.



Example of a gaze path.

How to enable

In the gaze replay window, you can enable the Gaze smoothing filter by selecting the "Gaze smoothing filter" checkbox under "*Visualizations*" in the *Analysis Window*. Please note that the filter is used exclusively for visualization purposes. Turning it on will not change the results of the fixation filter or any exported data.



Screenshot of the user interface to activate the Gaze smoothing filter.

Methods

To compute the smoothing filter at any point at time t along the gaze path, the following steps are applied:

1. The neighboring gaze points lying in a window of $[t - 80 \text{ ms}, t + 80 \text{ ms}]$ around the current points are selected. If missing gaze points are present in this interval, fewer points are considered.
2. Weighted averages are computed for the x and y coordinates of all points in the window, using a Hamming window of length 160 ms as weights.
3. The result of the weighted average gives the (x,y) coordinates of the smoothed point.

The smoothing filter makes the gaze path less jagged and mitigates outliers that might be caused by the signal noise. The reason for using a Hamming window is that the gaze points closest in time to the current point give a larger contribution to the result, making the smoothing more stable.

